Independent audit of the projected one-loop Boussinesq determinant: operator domain, intrinsic tail divergence, and conditional fixed-P invariants

Quantum Boussinesq solitons project — Independent audit for Gustav Delius

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Abstract

We report an independent mathematical and numerical audit of the projected one-loop determinant package `phase2/det_projected_oneloop.tex` and its scripts `det_geometry.py`, `det_spectrum.py`, and `det_charges.py`. All analytical identities are re-derived from the fixed good-Boussinesq Lagrangian without reusing package code paths, and all numerical quadratures and limits are recomputed at 70-digit working precision in `phase2/scripts/audit_det_projected.py` using independent coordinate transforms, closed-form dilogarithms, and Richardson extrapolations. **(c)** Every mathematical formula displayed in `det_projected_oneloop.tex` is confirmed; no displayed mathematical claim is refuted. We prove that the linear box divergence of the configuration-space kinetic norms of shift and velocity/width paths is *intrinsic* to the velocity-dependent topological jump $M(v) = 2\sqrt{1-v^2}$ of the potential formulation on $\mathbb{R}$, rather than an artefact of the auxiliary symplectic completion or the centred shift gauge. We confirm the "heavy-mode shorthand error" in `oneloop_phase.tex` and derive its exact analytical residual $4\cos\alpha\,\gamma(\alpha)(f_-^{\mathrm{sp}})'(\xi)$. We separate the unconditional Legendre algebra in $E_1|_P = -L_1$ from the conditional equal-mode-count cutoff endpoint B_Λ and the vacuum tadpole subtraction C_Λ. Finally, we qualify four tautological assertions and two text-to-script gaps in the package's verification scripts and supply non-circular replacements. **(d)** The full constrained functional determinant, nonzero-frequency spectral completeness, and absolute two-soliton phase normalization remain open.

1 Scope, conventions, and runtime verification

Claim labels follow `AGENTS.md`: **(a)** proved in the literature, **(b)** conjectured or heuristic in the literature, **(c)** derived or verified in this project, and **(d)** a guess or unestablished physical identification. No external references are invented; only local repository files were read in this session. We use the fixed good-equation conventions of `AGENTS.md`:

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \phi_x = 2(\log\tau)_{xx}, \qquad \mathcal{L} = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \phi_x^3 - \tfrac{1}{2}\phi_{xx}^2, \tag{1}$$

with $\pi = \phi_t$, $\Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\phi$, $k = \sin\alpha = \sqrt{1-v^2}$, $v = \epsilon\cos\alpha$, $M = 2k$, $\Pi = -2vk$, and $P = \frac{2}{3}vk^3$.

Package execution and runtimes (Point 6). All three package scripts and the independent audit script were executed from the repository root and finished by printing `ALL CHECKS PASSED` **(c)** :

Script (`phase2/scripts/`)	Precision	Cached JSON	Fresh run	Terminal status
`det_geometry.py`	55 dps	43.55 s	17.59 s	`ALL CHECKS PASSED`
`det_spectrum.py`	60 dps	1.91 s	0.68 s	`ALL CHECKS PASSED`
`det_charges.py`	65 dps	69.38 s	22.59 s	`ALL CHECKS PASSED`
`audit_det_projected.py`	70 dps	—	68.63 s	`ALL CHECKS PASSED`

All numerical outputs and SHA-256 digests of the audited files are saved in `phase2/scripts/audit_det_projected_results.json`.

2 Fluctuation operator, endpoint variations, and zero modes (Point 1)

2.1 First variation and spatial boundary concomitant

Expanding $\phi \mapsto \phi + \delta\phi$ in (1) and integrating by parts in (t, x) yields the exact identity **(c)** (`audit_det_projected.py`, check 1):

$$\delta\mathcal{L} = -\big[\phi_{tt} - \phi_{xx} + \phi_{xxxx} + 3(\phi_x^2)_x\big]\delta\phi + \partial_t(\pi\delta\phi) + \partial_x(A\delta\phi - u_x\delta\phi_x), \qquad A = -u + 3u^2 + u_{xx}. \tag{2}$$

On the classical soliton $u_0(\xi) = \frac{1}{2}k^2\,\mathrm{sech}^2(k\xi/2)$ ($\xi = x - X - vt$), both $A = (k^2-1)u_0 = -v^2u_0$ and $u_{0,x}$ decay exponentially as $x \to \pm\infty$. Consequently, time-independent bounded variations $\delta\phi(\pm\infty) \neq 0$ produce zero spatial boundary term in $\delta S|_{\text{space}}$. Similarly, local conservation $\pi_t = -\partial_x A$ implies $\dot{M} = [\pi]_{-\infty}^{+\infty} = 0$ and $\dot{\Pi} = [-A]_{-\infty}^{+\infty} = 0$ for smooth histories with $\pi(\pm\infty) = u(\pm\infty) = u_{xx}(\pm\infty) = 0$ **(c)** .

Qualification on `det_geometry.py` lines 25–33. Whereas Eq. (7) of `det_projected_oneloop.tex` is correct, lines 26 and 33 of `det_geometry.py` only assert the tautology $\partial_x(-u + 3u^2 + u_{xx}) = -u_x + 6uu_x + u_{xxx}$ (twice, with opposite signs and an unused symbol `p`) without testing the boundary form $A\delta\phi - u_x\delta\phi_x$ against $\delta\mathcal{L}$ **(c)** . Check 1 of `audit_det_projected.py` verifies the full identity (2).

2.2 Co-moving quadratic action and static kernel exhaustiveness

Writing $\phi(x,t) = \Phi(\xi; v, C) + \eta(\xi, t)$ with $\Phi(\xi; v, C) = C + k\tanh(k\xi/2)$, the second variation of (1) in co-moving coordinates is **(c)** (`audit_det_projected.py`, check 1)

$$\mathcal{L}_2 = \tfrac{1}{2}(\eta_t - v\eta_\xi)^2 - \tfrac{1}{2}(1 - 6u_0)\eta_\xi^2 - \tfrac{1}{2}\eta_{\xi\xi}^2 = -\tfrac{1}{2}\eta\mathcal{D}_v\eta + \partial_t(\cdots) + \partial_\xi(\cdots), \tag{3}$$

where

$$\mathcal{D}_v = (\partial_t - v\partial_\xi)^2 + \mathcal{L}, \qquad \mathcal{K}_v = \mathcal{L} + v^2\partial_\xi^2 = \partial_\xi^4 - k^2\partial_\xi^2 + 6\partial_\xi(u_0\partial_\xi) = -\partial_\xi\mathcal{H}_v\partial_\xi, \tag{4}$$

with $\mathcal{H}_v = -\partial_\xi^2 + k^2 - 6u_0$, and Lagrange–Green concomitant $f\mathcal{K}_v g - g\mathcal{K}_v f = \partial_\xi B_v(f,g)$, $B_v(f,g) = fg''' - f'g'' + f''g' - f'''g + (6u_0 + v^2 - 1)(fg' - gf')$. For monochromatic co-moving modes, differentiating $e^{-i\nu t}f(\xi)$ symbolically gives $\mathcal{D}_v(e^{-i\nu t}f) = e^{-i\nu t}(\mathcal{K}_v + 2iv\nu\partial_\xi - \nu^2)f$ **(c)** .

In the polynomial variable $y = \tanh(k\xi/2)$, $\mathcal{K}_v f = -\partial_\xi(\mathcal{H}_v f') = 0$ with $f'(\pm\infty) = 0$ integrates to $\mathcal{H}_v f' = 0$. One solution is $h_1(\xi) = u_0'(\xi) = -\frac{1}{2}k^3 y(1-y^2)$. Constructing the second linearly independent solution of $\mathcal{H}_v h = 0$ explicitly by reduction of order in SymPy (`audit_det_projected.py`, check 1) gives

$$h_2(y) = h_1(y)\int \frac{k^3\,\mathrm{d}y}{2(1-y^2)h_1(y)^2} = \frac{k\big(15y^4 - 18y^2 - 8y^2(1-y^2)\ln|y| + 3\big)}{8y(1-y^2)}, \tag{5}$$

which has constant Wronskian $W(h_1, h_2) = \frac{1}{2}k^4 \neq 0$ and diverges at $y \to \pm 1$ as $(1-y^2)^{-1} = \cosh^2(k\xi/2) \sim \frac{1}{4}e^{k|\xi|}$ **(c)** . Hence every solution of $\mathcal{K}_v f = 0$ with decaying derivative satisfies $f' = c_1 u_0'$, proving that $\ker_{\text{bdd}}\mathcal{K}_v = \mathrm{span}\{1, u_0(\xi)\}$. Moreover, differentiating the profile ODE $\mathcal{H}_v u_0' = 0$ with respect to v yields $\mathcal{H}_v(\partial_v u_0) = 2vu_0$ and hence $\mathcal{K}_v\Phi_v = -2vu_0'$ and $\mathcal{D}_v(\Phi_v - tu_0) = 0$ identically **(c)** .

3 Collective measure, projector tails, and intrinsic norm divergence (Point 2)

3.1 Rank-two three-tangent matrix and four-mode completion

Ordering the $t = 0$ soliton-family tangents as (v, X, C), namely $Z_v = (\Phi_v, -u_0 - v\partial_v u_0)$, $Z_X = (-u_0, vu_0')$, and $Z_C = (1, 0)$, their symplectic pairings $\Omega(Z_A, Z_B) = \int(\delta_A\pi\,\delta_B\phi - \delta_B\pi\,\delta_A\phi)\,\mathrm{d}\xi$ form the 3×3 skew-symmetric matrix (6) of rank two with null vector $\Pi_v\partial_X - P_v\partial_C$ **(c)** (`audit_det_projected.py`, check 2):

$$W_3 = \begin{pmatrix} 0 & P_v & \Pi_v \\ -P_v & 0 & 0 \\ -\Pi_v & 0 & 0 \end{pmatrix}, \qquad P_v = \frac{\mathrm{d}P}{\mathrm{d}v} = \frac{2}{3}k(1-4v^2), \quad \Pi_v = \frac{\mathrm{d}\Pi}{\mathrm{d}v} = \frac{2(2v^2-1)}{k}. \tag{6}$$

Appending the localized momentum direction $Z_r = (0, w)$ with $w(\xi) = \frac{3}{k}u_0 - \frac{15}{2k^3}u_0^2$ ($\int w\,\mathrm{d}\xi = 1$, $\int wu_0\,\mathrm{d}\xi = 0$, $\int w\Phi_v\,\mathrm{d}\xi = 0$ by parity) yields the invertible 4×4 matrix W_4 with $\det W_4 = P_v^2$ and Liouville density $\sqrt{\det W_4} = |P_v|$ away from the cusp $|v| = 1/2$ **(c)** . Independent 70-digit quadratures in `audit_det_projected.py` using the compactified coordinate $u = \tanh(k\xi/2) \in (-1, 1)$ at $v \in \{0.8, -0.7, 0.3, 0.6, -0.4\}$ match W_4 to $< 3\times 10^{-68}$ **(c)** .

3.2 Explicit field-space projector tail formula (qualification on line 297)

Line 297 of `det_projected_oneloop.tex` labels as "**(c)** (`det_geometry.py`)" the statement that the symplectic projector $\mathcal{P}\eta = Z_A(W_4^{-1})^{AB}\Omega(Z_B, \eta)$ does not preserve zero endpoint variations and can create both constant and opposite-sign tails. However, `det_geometry.py` (lines 64–77) tests only a 6×6 constant matrix and never evaluates $\mathcal{P}\eta$ on fields **(c)** . Inverting W_4 explicitly gives (`audit_det_projected.py`, check 2)

$$W_4^{-1} = \begin{pmatrix} 0 & -P_v^{-1} & 0 & 0 \\ P_v^{-1} & 0 & 0 & -P_v^{-1}\Pi_v \\ 0 & 0 & 0 & 1 \\ 0 & P_v^{-1}\Pi_v & -1 & 0 \end{pmatrix} \implies (\mathcal{P}\eta)_\phi(\pm\infty) = b_r \pm \frac{v}{kP_v}b_X, \tag{7}$$

where $b_r = \int w\eta_\phi\,\mathrm{d}\xi$ and $b_X = \int(vu_0'\eta_\phi + u_0\eta_\pi)\,\mathrm{d}\xi$. For the localized Schwartz-class perturbation $\eta = (w, 0)$, $b_r = \int w^2\,\mathrm{d}\xi = \frac{3}{7}k \neq 0$ and $b_X = 0$, producing a pure constant tail $(\mathcal{P}\eta)_\phi(\pm\infty) = \frac{3}{7}k$. For $\eta = (0, u_0)$, $b_r = 0$ and $b_X = \frac{2}{3}k^3 \neq 0$, producing an opposite-sign tail $(\mathcal{P}\eta)_\phi(\pm\infty) = \pm\frac{2vk^2}{3P_v}$ **(c)** . This supplies the missing field-space verification.

3.3 Why the kinetic norm divergence is intrinsic (Point 2)

For $\Phi(\xi; v, C) = C + k\tanh(k\xi/2)$, the velocity tangent $\Phi_v(\xi) = -\frac{v}{k}[\tanh(k\xi/2) + \frac{k\xi}{2}\operatorname{sech}^2(k\xi/2)]$ approaches $\mp\frac{v}{k}$ as $\xi \to \pm\infty$. In `audit_det_projected.py` (check 2) we evaluate both the leading box divergence on $[-R, R]$ and the exact $O(1)$ constant:

$$\int_{-R}^{R}\Phi_v(\xi)^2\,\mathrm{d}\xi = 2R\frac{v^2}{k^2} + \frac{2(\pi^2-6)}{9}\frac{v^2}{k^3} + O(R^2e^{-kR}), \tag{8}$$

verified numerically at $R = 180/k$ to $< 10^{-60}$ **(c)** .

Intrinsic vs. completion-artefact analysis. The linear divergence in (8) is **intrinsic** to the configuration-space potential formulation of the Boussinesq one-soliton on $\mathbb{R}$, and is **neither** an artefact of the auxiliary completion $Z_r = (0, w)$ **nor** an artefact of the centred shift coordinate C:

1. The auxiliary completion $Z_r = (0, w)$ is purely a localized momentum perturbation ($\delta\phi = 0$, $\delta\pi = w \sim \mathrm{sech}^2$) and does not enter the configuration-space tangents $\Phi_C = 1$ or Φ_v at all **(c)** .

2. Regardless of how the shift coordinate C is parametrized relative to v, any one-soliton potential profile on $\mathbb{R}$ has asymptotic kink jump equal to the topological charge $\phi(+\infty, t) - \phi(-\infty, t) = \int_{-\infty}^{\infty} u_0 \, \mathrm{d}x = M(v) = 2\sqrt{1 - v^2}$. Along any time-modulated collective path $(v(t), C(t))$ without compensating radiation tails, the endpoint velocities $\dot\phi_\pm = \dot C \mp \frac{v}{k}\dot v$ obey $\dot\phi_+ - \dot\phi_- = \dot M = -\frac{2v}{k}\dot v$. Even if one couples $\dot C = \lambda \dot v$ (for example $\lambda = \pm v/k$ to anchor one spatial endpoint at zero), the kinetic norm density slope is **(c)** (`audit_det_projected.py`, check 2)

$$\lim_{R\to\infty} \frac{1}{2R} \int_{-R}^{R} \tfrac{1}{2}\dot\Phi^2 \, \mathrm{d}\xi = \tfrac{1}{4}(\dot\phi_+^2 + \dot\phi_-^2) = \tfrac{1}{2}\left(\dot C^2 + \tfrac{1}{4}\dot M^2\right) = \frac{1}{2}\left(\lambda^2 + \frac{v^2}{k^2}\right)\dot v^2 \geq \frac{v^2}{2k^2}\dot v^2 > 0 \quad (9)$$

   for all $\lambda \in \mathbb{R}$ whenever $v \neq 0$ and $\dot v \neq 0$. The centred choice $\lambda = 0$ actually *minimizes* the divergence coefficient; anchoring one endpoint ($\lambda = \pm v/k$) doubles it.

3. In phase space, by contrast, the stationary soliton has exponentially localized canonical momentum $\pi_0 = -v u_0$, so its velocity derivative along the stationary family $\partial_v \pi_0 = -u_0 - v\partial_v u_0$ is exponentially localized. That is why the instantaneous symplectic form Ω is finite on (Z_v, Z_X, Z_C, Z_r), whereas promoting $v(t)$ and $C(t)$ to independent time-dependent configuration-space path-integral coordinates with fixed-endpoint radiation fields is obstructed by $\dot M = [\pi]_{-\infty}^{+\infty} = 0$ **(c)** .

4 Heavy-mode shorthand error and spectrum audit (Point 4)

For real $q \neq 0$, the continuum potential modes $f_q(\xi) = \partial_\xi[e^{iq\xi}(1 + Ae^{k\xi})/(1 + e^{k\xi})]$ satisfy the co-moving pencil $(\mathcal{K}_v + 2iv\nu\partial_\xi - \nu^2) f_q = 0$ with $\nu = |q|\sqrt{1 + q^2} - vq$ and $|A| = 1$, verified in `audit_det_projected.py` (check 3) at independent test points to $< 10^{-66}$ **(c)** .

On the heavy branch $\alpha \in (\pi/3, \pi/2)$, the secondary KP-triple wavenumbers $K_\mp = \sin(\alpha \mp \pi/3) \in (0, k)$ satisfy $K_- + K_+ = k$ and yield localized exponentially growing and decaying solutions $\eta_\mp(\xi, t) = e^{\pm\epsilon\gamma t} f_\mp^{\mathrm{sp}}(\xi)$ with $\gamma(\alpha) = \sqrt{3}(\sin^2\alpha - 3/4) > 0$ and $f_\mp^{\mathrm{sp}}(\xi) = \partial_\xi[e^{K_\mp \xi}/(1 + e^{k\xi})]$.

Confirmation of the heavy-mode shorthand error (Point 4). In `oneloop_phase.tex` (line 221), both growth rates $\pm\gamma(\alpha)$ were written with a single spatial profile $e^{\pm\gamma t}\partial_\xi^2[e^{K_{cb}\xi}/(1 + e^{k_1\xi})]$ ($K_{cb} = K_-$). (Notably, the accompanying script `oneloop_good_phonon_scattering.py` at line 267 actually looped over `((+1, K_cb), (-1, K_ac))`, contradicting its own comment at line 265.) Because the co-moving operator $\mathcal{K}_v - 2v\lambda\partial_\xi + \lambda^2$ contains the Doppler-odd term $-2v\lambda\partial_\xi$, reversing the rate $\lambda = \epsilon\gamma \mapsto -\epsilon\gamma$ requires reflecting $\xi \mapsto -\xi$, which sends $K_- \mapsto k - K_- = K_+$. Pairing f_-^{sp} with the opposite rate $-\epsilon\gamma$ leaves the exact analytical residual **(c)** (`audit_det_projected.py`, check 3):

$$\left(\mathcal{K}_v - 2v(-\epsilon\gamma)\partial_\xi + \gamma^2\right) f_-^{\mathrm{sp}}(\xi) = 4v\epsilon\gamma \, \partial_\xi f_-^{\mathrm{sp}}(\xi) = 4\cos\alpha \, \gamma(\alpha) \, (f_-^{\mathrm{sp}})'(\xi). \quad (10)$$

Formula (10) vanishes only at $v = 0$ ($\alpha = \pi/2$), is strictly independent of $\epsilon = \pm 1$ (since $v\epsilon = \cos\alpha$), and at $\alpha = 1.2$, $\xi = 0.37$ evaluates to $0.00733252608902684591052563782 0\ldots$, matching the 60-digit value recorded in `det_spectrum_results.json` to $< 10^{-65}$ **(c)** .

5 Fixed-momentum correction $E_1|_P = -L_1$ and the $-2/\pi$ limit (Point 3)

5.1 Independent high-precision recomputation and cutoff asymptotics

For $0 < \alpha < \pi/3$, let $\varphi_r = (\alpha+2\pi/3, \alpha-2\pi/3, \alpha)$ and $c_r = (1, 1, -2)$. In `audit_det_projected.py` (check 4) we recompute the mode-sum charges $e(\alpha)$, $p(\alpha, \epsilon)$, the canonical momentum $p_{\rm can} = \mathrm{d}L_1/\mathrm{d}v$, and $E_1|_P = -L_1 = e - vp$ at 70-digit precision using hyperbolic quadrature $q = \sinh u$, matching Table 1 of `det_projected_oneloop.tex` to all displayed digits **(c)** :

α	$p(\alpha,+1)$	$p_{\rm can}(\alpha,+1)$	$E_1\|_P = -L_1(\alpha)$	$\Lambda = 10^{22}$ Richardson error
0.15	−0.157387066429280	−0.190035364892327	−0.001424902440969	2.43×10^{-69}
0.35	−0.265196583086379	−0.433826858902482	−0.017687351910675	7.30×10^{-69}
0.50	−0.309798364829473	−0.603389892153812	−0.050091498992296	1.25×10^{-68}
0.85	−0.311827472658892	−0.962723079254963	−0.223358039968532	2.51×10^{-68}

Expanding the regulated energy trace at finite cutoff Λ yields the exact asymptotic remainder **(c)** (`audit_det_projected.py`, check 4):

$$\Delta E_{\rm reg}(\Lambda) - e(\alpha) := \frac{1}{4\pi}\int_{-\Lambda}^{\Lambda} \omega\theta'\,\mathrm{d}q - B_\Lambda + \frac{3k}{\pi}(\sqrt{1+\Lambda^2} - 1) - e(\alpha) = \frac{3\sin\alpha}{4\pi\Lambda} + O(\Lambda^{-3}). \quad (11)$$

At $\Lambda = 10^{22}$, the leading term $\frac{3\sin\alpha}{4\pi\Lambda}$ equals 3.56757×10^{-24}, 1.14454×10^{-23}, and 1.79355×10^{-23} at $\alpha = 0.15, 0.50, 0.85$, accounting for 64 digits of the `cutoff_error` reported in `det_charges_results.json`. Two-point Richardson extrapolation between $\Lambda = 10^{22}$ and 2×10^{22} removes the $O(\Lambda^{-1})$ tail and reduces the residual below 2.6×10^{-68} **(c)** .

5.2 Exact symbolic small-amplitude expansion

Whereas `det_projected_oneloop.tex` (line 432) attributes its small-α series to "analytic differentiation," `det_charges.py` (lines 80–85) actually used numerical finite differences (`mp.diff`). In `audit_det_projected.py` (check 4) we derive the series symbolically in SymPy from the exact rational-trigonometric derivative

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\ln\left|\frac{\tan((\alpha+2\pi/3)/2)}{\tan((\alpha-2\pi/3)/2)}\right| = \frac{4\sqrt{3}\cos\alpha}{3 - 4\sin^2\alpha}, \quad (12)$$

and $\mathrm{d}(E_1|_P)/\mathrm{d}\alpha = \epsilon\sin\alpha\, p_{\rm can}(\alpha, \epsilon)$, obtaining **(c)**

$$p_{\rm can}(\alpha, \epsilon) = -\frac{4\epsilon}{\pi}\alpha + \frac{8\epsilon}{9\pi}\alpha^3 - \frac{8\epsilon}{45\pi}\alpha^5 + O(\alpha^7), \quad (13)$$

$$E_1|_P(\alpha) = -\frac{4}{3\pi}\alpha^3 + \frac{14}{45\pi}\alpha^5 + O(\alpha^7), \quad (14)$$

$$\frac{E_1|_P(\alpha)}{|P_0(\alpha)|} = -\frac{2}{\pi} - \frac{23}{15\pi}\alpha^2 + O(\alpha^4) \underset{\alpha\to 0^+}{\longrightarrow} -\frac{2}{\pi} = -0.63661977236758134307553505\ldots. \quad (15)$$

Four-point Neville–Richardson extrapolation on $\alpha_k = 10^{-5}/2^k$ $(k = 0, 1, 2, 3)$ recovers $-2/\pi$ to 1.47×10^{-45} **(c)** .

5.3 Which ingredients are conditional vs. unconditional? (Point 3)

1. **Unconditional (choice between the two published charge pairs at fixed L_1):** Given $E_0'(\alpha) = v(\alpha)P_0'(\alpha)$ with $P_0'(\alpha) \neq 0$, first-order inversion of $P = P_0(\alpha) + \hbar\tilde{p}(\alpha)$ in $E = E_0(\alpha) + \hbar\tilde{e}(\alpha)$ gives $E_1|_P = \tilde{e} - v\tilde{p}$. Because the mode-sum pair (e, p) and the canonical pair $(e_{\rm can}, p_{\rm can}) = (vp_{\rm can} - L_1, \mathrm{d}L_1/\mathrm{d}v)$ are related by a tangent shift $(\delta e, \delta p) = (v\delta p, \delta p)$

sharing the same Legendre function $L_1 = vp - e = vp_{\rm can} - e_{\rm can}$, the equality $e - vp = e_{\rm can} - vp_{\rm can} = -L_1$ holds *unconditionally as an algebraic identity* **(c)** . Thus $E_1|_P$ does *not* depend on which of the two charge pairs is physical (though two-soliton frame shifts $\delta_{\rm renorm} = -\sum_j p_j \partial_{P_j}\delta_0$ do depend on the choice of p_j).

2. **Conditional Ingredient A (the equal-mode-count cutoff endpoint B_Λ):** The continuum energy formula $e(\alpha)$ requires retaining the integration-by-parts boundary term $B_\Lambda = \frac{\Lambda\sqrt{1+\Lambda^2}}{4\pi}[\theta(\Lambda) - \theta(-\Lambda)] = \frac{3k}{2\pi}\Lambda + O(\Lambda^{-1})$. Omitting B_Λ leaves an uncancelled linear UV divergence $+\frac{3k}{2\pi}\Lambda$ after adding $\delta E_{\rm ct}/\hbar$ **(c)** .

3. **Conditional Ingredient B (the renormalisation scheme and projected determinant):** The limit $E_1|_P/|P_0| \to -2/\pi$ relies critically on the exact cancellation of the $O(\alpha)$ linear term in $L_1(\alpha) = \frac{1}{2\pi}[3\sin\alpha + \sum_{r=1}^{2}\sin\varphi_r(\cos\varphi_r - \cos\alpha)\ln|\tan(\varphi_r/2)|]$, where $+3\sin\alpha$ originates from the finite constant -1 in the vacuum normal-ordering subtraction $C_\Lambda = \frac{\hbar}{2\pi}(\sqrt{1+\Lambda^2} - 1)$ combined with B_Λ. If the renormalisation prescription or the uncomputed boundary/Schur determinant shifts the finite tadpole subtraction by $\delta E_{\rm fin} = 2\hbar c_{\rm fin}\sin\alpha$ ($c_{\rm fin} \neq 0$), then $E_1|_P$ acquires $+2c_{\rm fin}\sin\alpha$ and the small-amplitude ratio diverges as **(c)** (`audit_det_projected.py`, check 4)

$$\frac{E_1|_P(\alpha) + 2c_{\rm fin}\sin\alpha}{|P_0(\alpha)|} = \frac{3c_{\rm fin}}{\alpha^2} - \frac{2}{\pi} + O(\alpha^0). \tag{16}$$

Likewise, identifying $L_1(v)$ with the full one-loop effective action remains conditional **(d)** on constructing the projected functional determinant.

6 Two-soliton phases and audit of script circularities (Point 5)

In `audit_det_projected.py` (check 5) we recompute the two-soliton continuum integrals I_{12}, I_{21}, J_{12} at $(\alpha_1, \alpha_2) = (0.3, 0.6)$ across all four orientations $(\epsilon_1, \epsilon_2) \in \{\pm 1\}^2$ both by hyperbolic quadrature $q = \sinh u$ and via the closed-form Legendre χ_2 representation, agreeing to $< 2\times 10^{-69}$ and confirming $I_{12} + I_{21} = -\frac{\pi}{2}(\epsilon_1 + \epsilon_2)$ and $J_{12} - I_{12} = \frac{\pi}{4}(\epsilon_1 + \epsilon_2)$, as well as the bootstrap counterexample $F_{12} = \lambda(P_1 M_2 - P_2 M_1)$ **(c)** .

Audit of circularities, tolerances, and text-to-script gaps (Point 5). No loosened tolerances (10^{-42} to 10^{-58} throughout) or target-feeding-back were found in the numerical calculations. However, auditing the script code line by line exposed four symbolic/numerical tautologies ("a formula compared with itself") and two text-to-script gaps **(c)** :

1. `det_geometry.py` lines 26 & 33: both assert $\partial_x(-u + 3u^2 + u_{xx}) = -u_x + 6uu_x + u_{xxx}$ (with opposite signs) without testing the boundary variation $A\delta\phi - u_x\delta\phi_x$ in Eq. (7).

2. `det_spectrum.py` lines 19–20: compares $(-i\omega)^2 f + 2iv\omega f' + v^2 f''$ against $-\omega^2 f + 2iv\omega f' + v^2 f''$ without differentiating $e^{-i\omega t}f(x - vt)$. (Line 17 also substitutes t into the vacuum operator where every derivative term is separately zero.)

3. `det_charges.py` lines 24 & 63: asserts $(v\,\delta p - l) - v\,\delta p + l = 0$ symbolically and $(vp_{\rm can} - L_1) - vp_{\rm can} = -L_1$ numerically (where L_1 was defined as $vp - e$), testing $(A - B) - A = -B$.

4. `det_charges.py` line 27: defines `label_phase = d0 + g1*(-dp1) + g2*(-dp2)` and asserts it equals `d0 - dp1*g1 - dp2*g2` without expanding $\delta_0(P_1, P_2)$.

5. Text-to-script gaps: line 297 of `det_projected_oneloop.tex` cites `det_geometry.py` for field-space projector tail creation (not in that script), and line 432 claims analytic differentiation for the small-α Taylor series whereas `det_charges.py` lines 80–85 use numerical `mp.diff`. All six items are supplied by non-circular checks in `audit_det_projected.py`.

7 Verdict table and failures recorded

Package claim (`det_projected_oneloop.tex`)	Audit verdict	Independent audit evidence (`audit_det_projected.py`)
Classical profile, shift dictionary, and charges (1), Eq. (1) **(c)**	**Confirmed (c)**	Check 1–2: exact SymPy integration on $y = \tanh(k\xi/2)$
Quadratic operator $\mathcal{D}_v$, $\mathcal{K}_v = -\partial_\xi \mathcal{H}_v \partial_\xi$, and Green concomitant B_v, Eqs. (2)–(9) **(c)**	**Qualified (c)**	Formulas confirmed; tautologies in `det_geometry.py:26,33` and `det_spectrum.py:19` replaced in Check 1
Linear box divergence of shift and velocity/width kinetic norms, Eqs. (12)–(14) **(c)**	**Confirmed (c)**	Check 2: proved intrinsic to $M(v) = 2\sqrt{1-v^2}$ across all shift gauges; exact $O(1)$ constant derived in (8)
Bounded static kernel $\{1, u_0\}$ and velocity Jacobi mode $\eta_v = \Phi_v - t u_0$, Eq. (15) **(c)**	**Confirmed (c)**	Check 1: exact symbolic $\mathcal{D}_v \eta_v = 0$ and reduction-of-order growing partner $h_2(\xi)$ in (5)
Continuum modes, heavy-branch modes, and `oneloop_phase.tex` shorthand error, Eqs. (16)–(21) **(c)**	**Confirmed (c)**	Check 3: 70-digit residuals $< 10^{-66}$; exact wrong-rate formula $4\cos\alpha\,\gamma(f_-^{\rm sp})'$ derived in (10)
Rank-two W_3, auxiliary completion W_4, $\det W_4 = P_v^2$, and finite projector, Eqs. (22)–(25) **(c)**	**Qualified (c)**	Check 2: W_3, W_4 confirmed to $< 3 \times 10^{-68}$; missing field-space tail check supplied by (7)
Vacuum counterterm C_Λ, equal-mode-count endpoint B_Λ, and regulated $e, p, L_1, p_{\rm can}$, Eqs. (26)–(35) **(c)**	**Confirmed (c)** (conditional)	Check 4: hyperbolic quadrature and exact cutoff tail $\frac{3\sin\alpha}{4\pi\Lambda}$ in (11); Richardson error $< 3 \times 10^{-67}$
Fixed-momentum invariant $E_1\|_P = -L_1$ and small-amplitude limit $-2/\pi$, Eqs. (36)–(41) **(c)**	**Qualified (c)** (conditional)	Check 4: symbolic SymPy series replaces numerical `mp.diff`; conditional dependence on B_Λ and C_Λ vs. unconditional charge-pair invariance established
Two-soliton exchange constants I_{12}, J_{12} and bootstrap family F_{12}, Eqs. (42)–(46) **(c)**	**Confirmed (c)**	Check 5: closed-form χ_2 vs. hyperbolic quadrature to $< 2 \times 10^{-69}$
Full constrained functional determinant, completeness, and physical $E(P), S$ **(d)**	**Confirmed (d)** (open)	Instantaneous completion does not remove temporal tail or Schur-complement obstructions

Failures recorded.

1. *Confirmed prior-claim failure:* Pairing $f_-^{\rm sp}(\xi)$ with the opposite growth rate $-\epsilon\gamma$ in the shorthand of `oneloop_phase.tex` (line 221) fails the co-moving PDE with residual $|4\cos\alpha\,\gamma(f_-^{\rm sp})'(\xi)| = 0.007332526\ldots$ at $\alpha = 1.2$, $\xi = 0.37$ (`audit_det_projected.py`, check 3).

2. *Confirmed mathematical obstructions:* Inverting the three-tangent Wronskian W_3 fails ($\operatorname{rank} W_3 = 2$); treating $C(t)$ and $v(t)$ as independent configuration-space collective paths fails by the intrinsic linear box divergence (9); omitting the equal-mode-count endpoint B_Λ fails UV cancellation by $+\frac{3k}{2\pi}\Lambda$; shifting the finite tadpole counterterm by $c_{\rm fin} \neq 0$ fails the finite small-amplitude limit by $3c_{\rm fin}/\alpha^2$ (`audit_det_projected.py`, checks 2 & 4).

3. *Package script circularities and gaps:* Four tautological assertions (`det_geometry.py` lines 26 & 33, `det_spectrum.py` lines 19–20, and `det_charges.py` lines 24, 27 & 63) and two text-

to-script omissions (field-space projector tails and symbolic small-α series) were detected and replaced by genuine checks in `audit_det_projected.py`.

4. *Audit script development failures (honestly recorded, fixed without weakening checks):* (i) In the first run of `audit_det_projected.py`, the L^2 norm $\int_{-\infty}^{\infty} w(\xi)^2\,\mathrm{d}\xi$ was typed from mental arithmetic as $\frac{6}{7k}$ instead of $\frac{3}{7}k$ and failed exact symbolic equality; recomputing $\int w^2\,\mathrm{d}\xi = (6 - 12 + \frac{45}{7})k = \frac{3}{7}k$ repaired the assertion. (ii) In the second run of `audit_det_projected.py`, expanding the even integrand $\sin\alpha\, p_{\mathrm{can}}(\alpha)$ with `sp.series(..., a, 0, 7)` retained the $O(\alpha^6)$ term, so its integral contained an $O(\alpha^7)$ term and failed exact polynomial equality against the $O(\alpha^5)$ truncation; changing the integrand series order to `6` resolved the polynomial check. (iii) In the third run of `audit_det_projected.py`, the analytical coefficient of Λ^{-1} in (11) omitted the $-\frac{3\sin\alpha}{4\pi\Lambda}$ contribution from $\sqrt{1+\Lambda^2} - \Lambda$ in the raw integral, giving $\frac{9\sin\alpha}{8\pi\Lambda}$ instead of $(-\frac{3}{8} + \frac{12}{8} - \frac{3}{8})\frac{\sin\alpha}{\pi\Lambda} = \frac{3\sin\alpha}{4\pi\Lambda}$ and failing the 10^{-64} assertion; restoring that term (with `mp.workdps(110)` at $\Lambda = 10^{22}$) passed with no weakened tolerance. (iv) In the fourth run of `audit_det_projected.py`, 3-point Richardson extrapolation at $\alpha \in \{10^{-4}, 5\times10^{-5}, 2.5\times10^{-5}\}$ had $O(\alpha_1^2\alpha_2^2\alpha_3^2) \sim 1.56\times10^{-26} > 10^{-28}$; upgrading to 4-point extrapolation at $\alpha_k = 10^{-5}/2^k$ allowed tightening the tolerance from 10^{-28} to 10^{-40}.